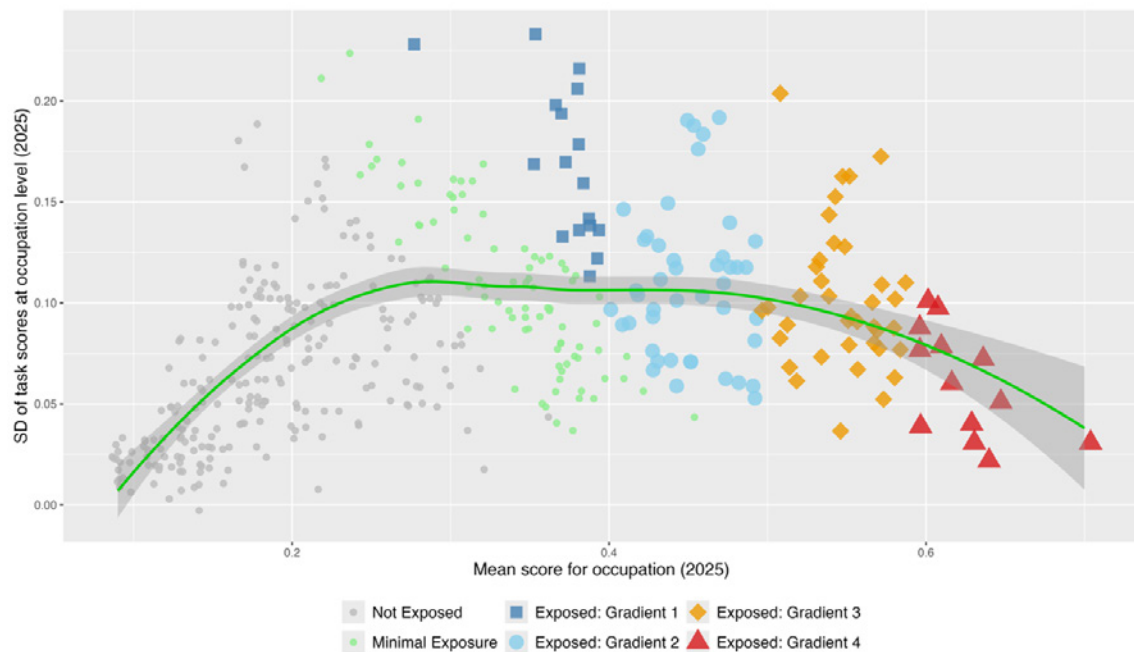


► 8 Adjusted Global Index of GenAI Exposure

To classify ISCO-08 occupations into varying levels of exposure to Generative AI (GenAI), we update the framework introduced in Gmyrek et al. (2023). Similarly to the original approach, we rely on two moments of distribution: occupational mean and standard deviation (SD) of task-level scores pertaining to a given 4-digit occupation. However, instead of using automation and augmentation potential as two extreme categories with the "big unknown" in between, we now adopt a more nuanced classification scheme that represents a spectrum of exposure, with occupations considered as exposed classified into four increasing gradients (Figure 16 and Table 5).

► Figure 16. AI Exposure Gradients: ISCO-08 4-digit level occupations



Note: Figure 16 presents all ISCO-08 4-digit occupations classified into GenAI exposure categories as per the criteria in Table 5. A detailed list of occupation-level means and SDs is available in the Appendix, Table A1. In addition, task-level scores within the overall hierarchy of ISCO-08 are available here: https://pgmyrek.github.io/2025_GenAI_scores_ISCO08/

Gradient 1 represents occupations with low overall GenAI exposure but significant variability across tasks (Low exposure, high task variability). While some tasks within these roles may have high automation potential, the occupation has many tasks that continue requiring human roles, aligning closely with the notion of augmentation in the old framework. Gradient 2 includes occupations with moderate GenAI exposure and a mix of highly exposed and minimally exposed tasks, resulting in uneven impacts where some tasks may be disrupted while others remain unaffected (Moderate exposure, high task variability). Gradient 3 captures occupations where a significant portion of tasks are consistently exposed to GenAI, signaling growing automation risks and requiring adaptation strategies for workers (Significant exposure, high task variability). Gradient 4 highlights occupations with the highest share of tasks exposed to potential GenAI-driven automation, and with a high consistency of this exposure across tasks (Highest exposure, low task variability). This gradient aligns closely with the notion of automation risk in our earlier classification system (Gmyrek et al., 2023). In addition, among occupations outside these four categories, we introduce a more granular distinction of Minimal Exposure, where some interaction with GenAI

may occur, but without significantly altering the nature of these roles, and the lowest category of Not Exposed occupations, where GenAI currently has no observable potential to automate tasks.

► **Table 5. GenAI Exposure Gradients: Definition and Interpretation**

GenAI Exposure	Definition (Mean & SD of Task Scores)	Interpretation
Exposed: Gradient 4 (Highest exposure, low task variability)	$\mu \geq 0.6$ and $\mu - \sigma \geq 0.5$	High and consistent GenAI exposure across tasks within the occupation. Most current tasks in these jobs have a high potential of automation, with little variability in task-level exposure.
Exposed: Gradient 3 (Significant exposure, high task variability)	$0.5 \leq \mu < 0.6$ and $\mu + \sigma \geq 0.5$	Above-moderate occupational exposure: even though some tasks remain less exposed, the overall potential of automation of the current tasks with GenAI is growing in these occupations.
Exposed: Gradient 2 (Moderate exposure, high task variability)	$0.4 \leq \mu < 0.5$ and $\mu + \sigma \geq 0.5$	Moderate occupational AI exposure, with high task-level variability. These occupations include a mix of some tasks that are exposed to GenAI and others not at risk, making the impact uneven.
Exposed: Gradient 1 (Low exposure, high task variability)	$\mu < 0.4$ and $\mu + \sigma \geq 0.5$	Low overall GenAI exposure at the occupational level, but high variability across tasks. Some tasks within these occupations have an elevated automation potential, even if the occupation as a whole remains strongly reliant on tasks that have a low potential of automation.
Minimal Exposure (Low exposure, moderate task variability)	$\mu < 0.5$ and $\mu + \sigma > 0.4$	Occupations with low GenAI exposure, where some tasks show moderate automation potential, but overall occupational exposure remains limited.
Not Exposed	Occupations that don't meet any of the above conditions.	Occupations where most tasks remain relatively unaffected by GenAI, with low task variability and a stable low mean exposure score.

This revised framework addresses some limitations of the earlier categorization system, particularly the interpretative challenges posed by the category, “the big unknown”. By expanding the exposure categories into a continuous gradient, the updated framework improves the readability and interpretability of results and enables us to stack employment estimates into the overall category of exposed occupations (Figure 17 in Section 10).¹⁸ It also captures in a more nuanced way how GenAI can impact occupations at varying levels of exposure, based on task-level variability and overall occupational risks. Furthermore, the framework's flexibility enhances its applicability to country-specific contexts, allowing policymakers to better understand the distribution of GenAI's potential impact and prioritize interventions for the most affected groups in the national contexts.

¹⁸ One of the limitations of the current approach is that neither ISCO-08 nor the Polish classification system includes information on the relative frequency or importance of tasks within occupations. As a result, all tasks are weighted equally when calculating exposure scores. Future research could refine this by incorporating task-level weights, for instance by mapping to occupational databases such as O*NET or by collecting self-reported data on task relevance directly from workers. A similar refinement could concern mapping the level of skills to individual tasks.

levels of exposure, highlighting the expanding abilities of GenAI technology to potentially automate more specialized tasks in professional and technical roles.

Overall, the revised global employment estimates of GenAI exposure are lower than in our 2023 paper, though still, one in four jobs globally (24%) has some degree of exposure, and one in three jobs in high-income countries. What is particularly striking is the continued association of gender with occupations with higher levels of exposure, especially in higher-income countries. Compared to 7.8% of female employment and 2.9% of male employment in high-income countries classified as having high automation potential in 2023, the current study increases these figures to 9.6% for females and 3.5% for males, respectively.

Notwithstanding these findings, we stress that such exposure does not imply the immediate automation of an entire occupation, but rather the potential for a large share of its current tasks to be performed using this technology. Whether this leads to the disappearance of an occupation or workforce replacement is a more complex question – one that will depend on the initial decision to adopt the technology, but also the extent to which individuals in these occupations are given opportunities to learn to work with these technologies and adapt to the evolving nature of their tasks. As argued by numerous economists, the key to unlocking the productivity benefits of GenAI lies not in the search of outright labour savings, but in the extent to which human expertise can be complemented with new technological capacities to reshape occupations and generate additional value-added (Acemoglu, 2024; Autor, 2024).

Ensuring such positive synergies necessitates an approach to technology's adoption at the workplace that is human-centered. With one in four workers across the world in an occupation with some degree of GenAI exposure, there is need to ensure that the transition is managed through social dialogue, both to minimize “technological unemployment” and the hardship associated with it, but also to ensure that for the majority who remain employed, the technology is implemented in a way that supports them in carrying out their jobs. Workers know their jobs best and can play an important role in the design, adaptation and use of the technology at the workplace, to the benefit of both working conditions and productivity.